

■ BLUESTONE

SEO Generation Pipeline

Source of Truth

Complete workflow reference for the AI-powered blog article pipeline

Version: 2026-09-29 | Client: BlueStone (India)

Website: blog.bluestone.com | Runner: AGY (Antigravity AI)

What Is This Pipeline? (Start Here)

If you are reading this document for the first time, this section explains what the pipeline does, who is involved, and how the pieces fit together — in plain English, with no assumed knowledge.

The Big Picture

BlueStone is an Indian fine jewellery brand (gold and diamonds). This pipeline automatically writes, publishes, and quality-checks SEO blog articles on **blog.bluestone.com**. Each article targets a specific keyword (e.g. 'gold ring designs for women 2026') and is designed to rank on Google India.

The entire process — from picking the next keyword to publishing a fully-formatted WordPress post with images, a product carousel, SEO metadata, and schema markup — is handled by an **AI agent** (called AGY) that follows this pipeline's rules autonomously. A human only needs to start the command and review the output.

Who Does What?

Actor	What They Do
You (the operator)	Run the pipeline command in your terminal. Review the final report. Approve or fix any issues.
Python Orchestrator (article_pipeline.py)	A script YOU run. It reads the keyword queue from an Excel file, builds a mega-prompt containing all context and rules, then launches the AI agent. After the agent finishes, it updates tracking files.
AI Agent (AGY)	The brain. It receives the mega-prompt and autonomously executes all 16 steps: researching competitors, writing the article, building a product carousel, publishing to WordPress, generating AI images, running live QA, and writing the final report. It does NOT need human input during its run.
WordPress (blog.bluestone.com)	The blog platform where articles are published. The agent talks to it via a REST API to create posts, upload images, set categories, and configure Yoast SEO.
Higgsfield AI	An external AI image generation service. The agent uses its CLI to create lifestyle/editorial images (called 'Type 3' images) for the hero and article body.
Google / Semrush	The agent scrapes Google India search results for the target keyword to understand what competitors are ranking for, then writes an article that covers the same topics AND fills gaps competitors missed.

How One Article Gets Made (The Journey)

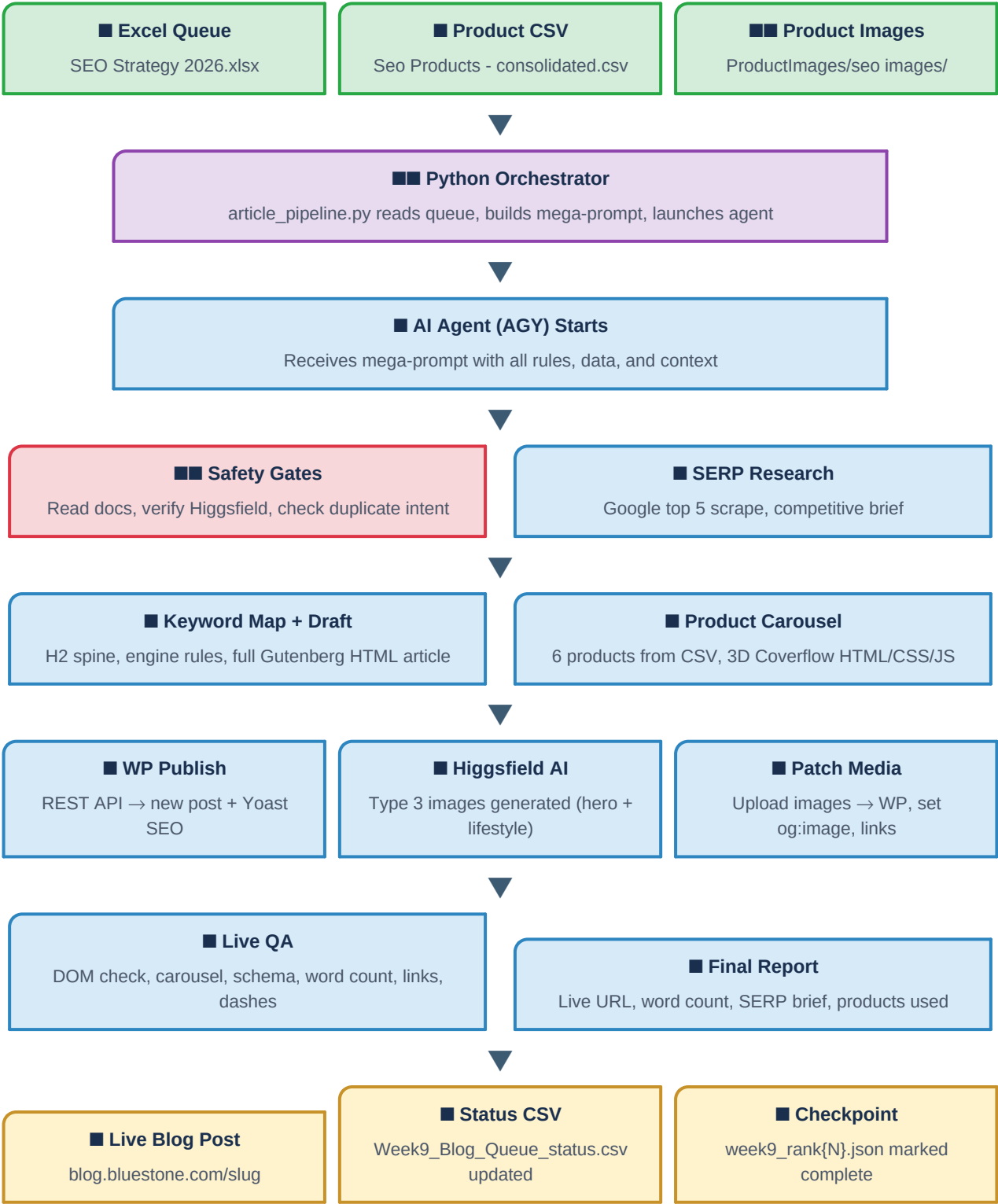
Here is the lifecycle of a single article, from start to finish:

- **1. You run the command:** `python3 -B scripts/article_pipeline.py produce --next ...` This tells the orchestrator to pick the next unfinished keyword from the Excel queue.
- **2. The orchestrator reads the keyword queue** (SEO Strategy 2026.xlsx) and finds the next row marked 'Action = New'. It loads the keyword, volume, difficulty, suggested slug, and article engine.

- **3. It builds a massive prompt** containing: the target keyword data, all editorial rules, all SOPs, image generation instructions, WordPress config, and any pre-fetched SERP data for the keyword.
- **4. It launches the AI agent (AGY)** with that prompt. From this point, the agent works autonomously.
- **5. The agent reads all docs** loaded into its context and verifies Higgsfield AI is accessible.
- **6. Safety gates:** The agent checks if this keyword/slug already has a live post on BlueStone's blog. If a duplicate exists, it stops safely without creating anything.
- **7. SERP research:** The agent searches Google India for the keyword, reads the top 5 ranking pages, and writes a competitive brief identifying table-stakes topics, common FAQs, and content gaps.
- **8. Article planning:** The agent maps keywords to H2 headings, picks the right article engine (Gift Guide / Buying Guide / Design Listicle), and outlines the visual concept.
- **9. Drafting:** The agent writes the full article in WordPress Gutenberg HTML format — including introduction, body sections, product recommendations, a 6-product carousel, FAQ section, and JSON-LD schema.
- **10. Publishing:** The agent creates a new WordPress post via the REST API, uploads the hero image, sets Yoast SEO fields (focus keyword, meta title, meta description), and assigns categories.
- **11. Image generation:** The agent uses Higgsfield AI to generate 2-3 lifestyle images (hero + in-body), then uploads them to WordPress and patches the article to include them.
- **12. Live QA:** The agent fetches the live published URL and verifies everything: carousel works, images load, FAQs render, schema parses, word count is good, no broken links, no forbidden dashes.
- **13. Status update:** The agent updates the tracking CSV, product rotation file, and checkpoint.
- **14. Final report:** The agent prints a summary with the live URL, word count, products used, SERP insights, and any issues found.

Visual Pipeline Flowchart

The diagram below shows the end-to-end flow from your terminal command to a live published article. Read it top-to-bottom. Green boxes are inputs, blue boxes are agent actions, gold boxes are outputs.



- **Green** = Inputs (files you provide)
- **Blue** = AI agent actions (fully autonomous)
- **Red** = Safety gates (can halt the pipeline)
- **Purple** = Python orchestrator (runs on your machine)
- **Gold** = Final outputs (what you get)

What Gets Updated After Each Article?

After the agent finishes and the article goes live, these files are updated to track progress:

File	What Gets Updated	How
output/Week9_Blog_Queue_status.csv	Rank row upserted with: live URL, WP post ID, status, word count, notes	Automatic (orchestrator)
output/checkpoints/week9_rank{N}.json	All 16 steps marked 'done' with timestamps. Status set to 'complete'	Automatic (agent + orchestrator)
output/product_rotation.json	SKUs used in this article added/updated to prevent overuse in next articles	Automatic (agent)
SEO Strategy 2026.xlsx (Week 9 sheet)	Target row updated with live URL, slug, and execution note	Semi-automatic (when safe) or manual
Manifest JSON (e.g. week9_69.json)	Run entry finalized with timing, exit code, outcome, and log paths	Automatic (orchestrator)

Quick Glossary

Term	What It Means
AGY	Antigravity AI — the AI coding agent that executes the pipeline steps
Orchestrator	The Python script (article_pipeline.py) that you run. It prepares data and launches AGY
Mega-prompt	A very large text prompt built by the orchestrator containing all context, rules, and data the agent needs
Checkpoint	A JSON file tracking which of the 16 steps are done. Enables crash recovery — just rerun the command
Type 1 image	Raw studio jewellery photos from BlueStone. Used as reference only, never published directly
Type 2 image	Styled product shots on plinths/fabric. Used in the mid-article product carousel
Type 3 image	AI-generated lifestyle/editorial images via Higgsfield. Used as hero and in-body visuals
Carousel	A 3D Coverflow interactive HTML widget showing 6 products with Buy Now links mid-article
SERP	Search Engine Results Page — the Google results for a keyword. Agent scrapes top 5 to inform content
Engine	Article template type: Gift Guide, Buying Guide (education), or Design Listicle
Rank	The priority number of a keyword row in the Excel queue (Rank 1 = highest priority)
PDP	Product Detail Page — the BlueStone product page URL linked from carousel cards and images
Yoast	WordPress SEO plugin. The agent sets focus keyword, meta title, and meta description via its API

■ Table of Contents

0. What Is This Pipeline? (Start Here)
 - The Big Picture
 - Who Does What?
 - How One Article Gets Made
 - Visual Pipeline Flowchart
 - What Gets Updated After Each Article
 - Quick Glossary
1. System Overview & Architecture
2. File & Directory Structure
3. The 16-Step Pipeline
 - Phase 1: Initialization & Safety Gates (Steps 1–4)
 - Phase 2: SERP Research & Strategy (Steps 5–6)
 - Phase 3: Content Architecture (Steps 7–8)
 - Phase 4: Content Production (Steps 9–10)
 - Phase 5: Publishing & Visual Production (Steps 11–13)
 - Phase 6: Quality Assurance & Finalization (Steps 14–16)
4. Content Engines (Gift Guide / Buying Guide / Design Listicle)
5. Image System (Types 1, 2, 3)
6. Carousel & Product Rules
7. WordPress Publishing & Yoast SEO
8. Checkpoint & Idempotency System
9. Mandatory Content Gates (Hard Rules)
10. CLI Reference & How to Run
11. Error Recovery Playbook
12. Monitoring & Status Tracking
13. Key Files Quick Reference

1. System Overview & Architecture

The BlueStone SEO Generation Pipeline is a fully-automated, AI-orchestrated system that produces complete, publication-ready SEO blog articles for blog.bluestone.com. Each article is generated end-to-end in a single AI agent session: from keyword research and competitor SERP analysis through drafting, product carousel assembly, WordPress publishing, Higgsfield AI image generation, live QA verification, and status tracking.

Architecture at a Glance

Layer	Component	Role
Input	SEO Strategy 2026.xlsx (Week 9 sheet)	Master keyword queue with rank, KD, volume, slug, theme
Input	Week9_Blog_Queue.csv	CSV fallback if XLSX is unavailable
Input	Seo Products - consolidated.csv	Product catalog with PDP URLs
Input	ProductImages/	Raw (Type 1) & styled (Type 2) jewellery photos
Orchestrator	article_pipeline.py	Python script that loads rows, builds prompts, launches AGY agent, finalizes checkpoints
Orchestrator	article_checkpoint.py	Atomic JSON checkpoint manager (16 steps)
Orchestrator	serp_competitor_intel.py	Pre-fetches Semrush SERP data per keyword
AI Agent	AGY (Antigravity)	Executes all 16 steps: research → draft → publish → QA
External	WordPress REST API	Publishes post, sets Yoast SEO fields, uploads media
External	Higgsfield CLI (nano_banana_pro)	Generates Type 3 lifestyle/concept images
External	Google Search / Semrush	SERP intelligence (top 5 competitor URLs)
Output	Published blog post	Live article on blog.bluestone.com
Output	Week9_Blog_Queue_status.csv	Terminal status tracker (rank, URL, WP post ID)
Output	Manifest JSON	Full run audit trail with timing and outcomes
Output	Checkpoint JSON (per rank)	Atomic step state for idempotent reruns

Design Principles

- **Single continuous agent session:** All 16 steps run in one AGY invocation. This preserves full context across steps (competitor research directly informs the draft; product selection informs image prompts).
- **Atomic idempotency:** Every step is checkpointed atomically. If the agent crashes, restarting the same command skips completed steps and resumes from the last pending one. WordPress post creation is strictly once.
- **SERP-first strategy:** The pipeline fetches and scrapes the top 5 Google India ranking pages BEFORE drafting. This ensures every article covers what Google currently rewards and differentiates on content gaps.

- **New posts only:** The pipeline only generates posts with Action=New. It never patches, optimizes, or deletes existing BlueStone blog posts.
- **Zero-price policy:** No prices appear in any article, carousel, or image. No competitor criticism. No fake statistics.

2. File & Directory Structure

The workspace root is the folder **seo final 2026**. Every path below is relative to this root.

File / Directory	Purpose
SEO Strategy 2026.xlsx	Master strategy workbook — Week 9 sheet is the primary queue
HANDOFF.md	Project context, state handoff, quick-start instructions
docs/PIPELINE_ARCHITECTURE.md	Full architecture reference (this PDF is the definitive version)
docs/SOP_ARTICLE_GENERATION.md	Primary execution runbook (Step 0 → Step 10)
docs/ARTICLE_WORKFLOW.md	Canonical hard rules for article production
docs/ARTICLE_ENGINES_PLAYBOOK.md	Engine selection: Gift Guide / Buying Guide / Design Listicle
docs/bluestone-blog-master-prompt-v5.md	Master drafting prompt template
docs/Blog-SEO-AEO-GEO-Checklist-v2.md	Pre-publish QA checklist (mandatory)
docs/HIGGSFIELD_IMAGE_GENERATION.md	Higgsfield image SOP, prompts, Image SEO table
cursor-rules/new-blogs-only.mdc	Rule: only create new WP posts, never optimize old ones
cursor-rules/carousel-seo-images.mdc	Carousel image sourcing & conversion rules
cursor-rules/type3-fair-skinned-indians.mdc	Type 3 diversity / skin tone guideline
cursor-rules/image-seo.mdc	Image SEO: alts, filenames, WebP, media library titles
cursor-rules/product-rotation-captions-education.mdc	Product rotation and caption rules
templates/eid_carousel_6_snippet.html	Official 3D Coverflow carousel HTML/CSS/JS template
scripts/article_pipeline.py	Main orchestrator — current production pipeline
scripts/article_checkpoint.py	Atomic checkpoint manager (16 steps)
scripts/week6_pipeline.py	Shared utilities (common module)
scripts/serp_competitor_intel.py	Semrush SERP intelligence helper
output/Week9_Blog_Queue.csv	Week 9 queue CSV (fallback source)
output/Week9_Blog_Queue_status.csv	Terminal status tracker (rank → URL, WP post ID)
output/product_rotation.json	SKU rotation state (tracks which products used per rank)
output/serp_cache/	Cached Semrush SERP results per keyword
output/checkpoints/	Per-rank atomic checkpoint JSONs
ProductImages/raw/	Type 1: original studio jewellery photos (reference only)
ProductImages/seo images/	Type 2: styled product shots for carousel (960×535 WebP)

File / Directory	Purpose
Seo Products - consolidated.csv	Product catalog with PDP URLs, SKU codes, categories

3. The 16-Step Pipeline

The pipeline orchestrator (`article_pipeline.py`) builds a single mega-prompt containing all context and rules, then launches the AGY AI agent. The agent executes all 16 steps synchronously within one session, marking each step in the checkpoint file as it goes.

Pipeline Flow Summary

Phase	Name	Steps	What Happens
Phase 1	Initialization & Safety	Steps 1–4	Read docs → Verify Higgsfield → Inspect row → Duplicate check
Phase 2	SERP Research	Steps 5–6	SERP intelligence (top 5 URLs) → Fact-check & compliance
Phase 3	Content Architecture	Steps 7–8	Keyword mapping & H2 spine → Article structure & visual concept
Phase 4	Content Production	Steps 9–10	Full article draft → Product curation & 3D carousel
Phase 5	Publishing & Visuals	Steps 11–13	WordPress publish → Higgsfield Type 3 generation → Media patching
Phase 6	QA & Finalization	Steps 14–16	Live QA verification → Status update → Final report

Phase 1: Initialization & Safety Gates

1

Read Required Documents

checkpoint: `read_docs`

The agent reads all editorial guidelines, SOPs, and cursor rules into context before doing any task work.

- `HANDOFF.md` — project state and active queue
- `docs/SOP_ARTICLE_GENERATION.md` — primary runbook
- `docs/ARTICLE_WORKFLOW.md` — canonical hard rules
- `docs/ARTICLE_ENGINES_PLAYBOOK.md` — engine selection
- `docs/HIGGSFIELD_IMAGE_GENERATION.md` — image SOP
- `docs/Blog-SEO-AEO-GEO-Checklist-v2.md` — QA checklist
- All 5 `cursor-rules .mdc` files

2

Verify Higgsfield AI Status*checkpoint: verify_higgsfield*

Before any image generation, the agent confirms Higgsfield credits, plan status, and CLI connectivity.

- Runs: higgsfield account status --json (or MCP balance tool)
- Reports plan name and remaining credit balance
- Pipeline STOPS here if status fails — no proceeding to image generation
- On auth failure: user must re-login or reconnect the Higgsfield connector

3

Inspect Target Row*checkpoint: inspect_row*

Validates the target keyword row from the Week 9 workbook loaded by article_pipeline.py.

- Data pre-loaded by orchestrator from SEO Strategy 2026.xlsx Week 9 sheet
- Validates: rank, primary keyword, slug, volume, KD, theme, engine, category fit
- Confirms Action = New (pipeline skips Done / Skip-Existing rows)
- Source: SEO Strategy 2026.xlsx Week 9 sheet OR Week9_Blog_Queue.csv fallback

4

Duplicate & Intent Gate*checkpoint: duplicate_check*

Checks if the target URL slug OR identical search intent already exists on blog.bluestone.com.

- Queries WordPress REST API for the target slug
- Searches existing posts for overlapping search intent
- If duplicate found: pipeline HALTS safely — no new post, no media, checkpoint marked skipped_existing_intent
- Override available: --allow-existing-intent flag (requires --restart for audit trail)
- This is a hard safety gate — cannot be bypassed accidentally

Phase 2: SERP Research & Strategy

5

SERP Intelligence**checkpoint:** *serp_intelligence*

Searches Google India for the primary keyword, scrapes the top 5 organic URLs, and synthesizes competitive brief. This step runs AFTER the duplicate check passes and BEFORE keyword mapping drafting.

- If Semrush data is cached in output/serp_cache/, pre-fetched URLs are used directly
- Otherwise: agent searches Google India and identifies top 5 organic results (excludes ads, featured snippets, PAA)
- For each of the 5 URLs: browses and reads full content, logs SERP_SCRAPED: position | url | domain | word_count | h2_count
- Extracts: page title, H2/H3 outline, key subtopics, FAQs answered, approximate word count
- Synthesizes SERP Competitive Brief containing:
 - a) Consensus topics (H2 themes in 3+ of top 5 pages — table stakes your article must cover)
 - b) Common FAQs (questions answered by 2+ competitors)
 - c) Content gaps (subtopics missing from all 5 pages — BlueStone differentiation opportunity)
 - d) Intent format (listicle / buying guide / long-form education / gift guide)
 - e) Average word count and depth benchmark
- Logs: SERP_BRIEF: <one-line summary of consensus topics and gaps>
- Uses brief directly to shape the H2 keyword map and article outline
- NEVER copies competitor wording — analysis only for topical coverage and differentiation

6

Fact-Check & Compliance**checkpoint:** *fact_check*

Verifies factual accuracy for buying and education topics. Browses authoritative sources when the keyword intent demands it.

- Triggered for: GST rates, BIS hallmarking, gold purity, gem grading, legal/tax claims
- Sources: BIS website, official government notifications, authoritative jewellery industry bodies
- Hard rules enforced: no fake statistics, no prices, no competitor criticism, correct festival year
- For gift guides: fact-check is lighter (no regulatory compliance required)
- All cited facts logged in the Final Report

Phase 3: Content Architecture

7

Keyword Mapping & H2 Spine

checkpoint: *keyword_map*

Maps the primary keyword plus all supporting keywords into a structured H2/H3 heading map and FAQ list

- Each H2 is labeled as: primary-backed / supporting-keyword-backed / competitor-structure-backed / serp-consensus-backed / intent-inferred
- Spine H2s (~70-80%): volume-backed primary and supporting keywords — placed higher in article
- Parity H2s (~20-30%): competitor structural buckets — placed lower, kept shorter
- Supporting keyword audit: irrelevant or polluted phrases explicitly REJECTED with reasons
- Engine selection applied: Gift Guide / Buying Guide / Design Listicle (from ARTICLE_ENGINES_PLAYBOOK.md)
- SERP brief consensus topics are mandatory inclusions in the H2 map

8

Article Structure & Visual Concept

checkpoint: *structure_visuals*

Defines the complete article outline including visual layout, section ordering, and image placement plan.

- Section ordering (MANDATORY): Intro → Body H2s → Carousel (mid-article) → More H2s → Related Guides → Conclusion → FAQs → JSON-LD
- Final Thoughts / Conclusion MUST appear BEFORE Frequently Asked Questions
- FAQs (5-7 visible Q&As) MUST be the final content section before JSON-LD schema
- Related Guides section (internal blog links) MUST appear between Conclusion and FAQs
- Visual plan: hero placement (featured image), in-body Type 3 image positions, carousel position

Phase 4: Content Production

9

Full Article Draft

checkpoint: *draft*

Writes the complete Gutenberg HTML article following the Master Prompt v5 and all content rules.

- Drafting guide: docs/bluestone-blog-master-prompt-v5.md
- Engine rules applied: Gift Guide / Buying Guide / Design Listicle structure and tone
- Gift guides: 5-6 named BlueStone recommendations with distinct gifting reasons, no prices
- Buying guides: explains calculations, buyer implications, regulatory context with citations
- Minimum internal links: 4+ internal BlueStone blog links + 1+ external authoritative link
- FAQs: 5-7 visible Q&A pairs in the HTML body (plus FAQPage JSON-LD schema)
- Author: Satyam (WP user ID 270271337) with byline 'By Satyam, BlueStone Editorial'
- ZERO em dashes, en dashes, or spaced hyphens anywhere in the article
- NO HTML tables — use structured bullet lists, cards, or styled PNG graphics instead
- NO duplicate sections — every H2 heading appears exactly once from intro to conclusion
- Strict Gutenberg block syntax with opening and closing comment tags on every block
- Every wp:paragraph block must explicitly enclose text in <p>...</p> tags

Product Curation & 3D Coverflow Carousel

checkpoint: *product_media*

Selects 5-6 products from the catalog and builds the official 3D Coverflow carousel.

- Product source: Seo Products - consolidated.csv (PDP links only, NO prices)
- SKU selection: 5-6 products matched to article intent, theme, and category
- Product rotation tracked in output/product_rotation.json (no SKU overuse)
- Carousel template: templates/eid_carousel_6_snippet.html (MUST use exactly this template)
- Carousel is a SINGLE unified wp:html block containing: <style> + <div class='bs-cf'> + <script>
- NEVER use unstyled wrappers (bs-cf-wrap, bs-cf-track) or split into multiple blocks
- Image source: Type 2 styled images from ProductImages/seo images/<Category>/<Name>.png
- Images converted to 960x535 WebP on stone plinths/draped fabric with warm lighting
- NEVER use plain white packshots for carousel cards
- Pre-flight check: EVERY carousel image URL verified to return HTTP 200 before insertion
- Every card includes: linked image (bs-cf-media) + product name + Buy now CTA (PDP URL)

Phase 5: Publishing & Visual Production

11

WordPress Publish & Yoast SEO

checkpoint: *publish_wordpress*

Creates the new WordPress post via REST API with complete Gutenberg content, then configures Yoast SEO fields.

- API auth: WP_USER + WP_APP_PASSWORD environment variables
- Base URL: <https://blog.bluestone.com/wp-json/wp/v2/>
- Post created with: Gutenberg HTML body, title, slug, categories, author ID 270271337
- WordPress creation is EXACTLY ONCE — reuses post if slug already exists from this run
- Never creates duplicate posts — stops safely if an untracked matching slug is found
- Yoast fields set: Focus Keyphrase, SEO Title (CTR-optimized), Meta Description (~155 chars)
- Featured image set as both featured_media and Yoast social image (og:image)
- Categories mapped by article intent — see WordPress Category Mapping table in Section 7

12

Higgsfield AI Image Generation (Type 3)

checkpoint: *generate_type3*

Generates lifestyle and conceptual images using the Higgsfield CLI nano_banana_pro model.

- Model: nano_banana_pro, aspect: 16:9, resolution: 2K, count: 1 per slot
- Slots: Hero image + 1-2 in-body lifestyle/conceptual images
- Style: analog grain, Kodak Portra color science, highlight halation, creamy bokeh, editorial color grading
- Max 2 concurrent generations — never exceed this limit
- On error: wait 45 seconds, then retry once. If still failing, step marked failed
- Extract image URL from result_url field in job JSON (NOT 'results' or 'output')
- Hero/lifestyle: image reference 1 = BP-PICS body_image, reference 2 = design angle
- If a SKU lacks body_image: select a different SKU — never generate people from packshots
- Product dimensions: product_height_mm and product_width_mm ONLY (not face/body measurements)
- FORBIDDEN in images: empty phones, screens, cards, boards, readable text, logos, product overlay
- Use physical props: flowers, diyas, ceramic cups, brass bowls, ribbons, wrapped gift boxes
- Jewellery material truth: BlueStone sells gold and diamond ONLY (no silver). Never describe gold/diamond SKUs as silver
- Output: WebP files with keyword-rich descriptive filenames

13

Media Patching & Social Metadata**checkpoint:** *patch_type3*

Uploads Type 3 images to WordPress, sets the featured hero, updates in-body image blocks, and configures social metadata.

- Uploads all Type 3 WebP images to WordPress Media Library via REST API
- Sets hero as featured_media AND Yoast social image (og:image)
- Patches in-body image blocks to use the new uploaded media IDs
- Every product image in body is wrapped in a direct PDP hyperlink
- For in-body Type 3: sets linkDestination: custom with href to PDP URL
- Image SEO: unique keyword-led alt text, descriptive media library titles, sizeSlug: full only
- CLEAN image captions: standard <figcaption> without extra classes
- Verifies no duplicate hero image appears in the article body

Phase 6: Quality Assurance & Finalization

14

Live QA Verification**checkpoint:** *live_qa*

Fetches the live published URL and runs a comprehensive DOM and metadata verification pass.

- Fetches the live published URL (cache-busted) and parses the HTML DOM
- CAROUSEL GATE: verifies .bs-cf, .bs-cf-stage, .bs-cf-card, .bs-cf-dots are present
- Verifies all 6 carousel card image URLs return HTTP 200 via HEAD requests
- If broken carousel detected: immediately replaces with eid_carousel_6_snippet.html template
- Single <h1> check (theme post title only — no injected content h1)
- Visible word count reported
- Heading counts: all planned H2s present in correct order
- All carousel Buy now links valid and pointing to correct PDPs
- 5-7 FAQ questions and answers visible in the HTML body
- FAQPage schema parses cleanly; BlogPosting schema images array populated
- 4+ internal links, 1+ external authoritative link
- og:image verified, canonical URL correct
- No forbidden dashes (em/en), no prices anywhere
- All images in WebP format with unique alt text and descriptive captions
- Author set to Satyam, categories correct
- Pass criteria: checkpoint detail result=passed

15

Status Update

checkpoint: *update_status*

Updates all tracking files to record the completed rank.

- Agent marks checkpoint `update_status` step done
- `article_pipeline.py` calls `upsert_status_record()` atomically with file-system locking
- `output/Week9_Blog_Queue_status.csv` updated (upsert by rank — never blind append)
- `output/product_rotation.json` updated (upsert by rank or SKU — never blind append)
- Status set to: published OR `skipped_existing_intent`
- Checkpoint status set to: complete with outcome and `completed_at` timestamp

16

Final Report

checkpoint: *final_report*

Agent writes a comprehensive final report covering all key metrics and audit information for the run.

- Live URL and WordPress post ID
- Article engine, author, categories
- Visible article word count
- Primary keyword and supplied volume/KD metrics
- Supporting keywords: which were used (with H2 placement) and which were rejected (with reasons)
- SERP intelligence: all URLs scraped (position, url, domain, word count, H2 count), consensus topics content gaps, average competitor word count
- H2 keyword map with SERP-informed labels (which H2s added based on competitor analysis)
- Competitor URL analyzed and key findings
- Factual sources cited
- Carousel media IDs and Type 3 media IDs
- Product SKUs used and product rotation update
- Higgsfield: plan, starting balance, jobs run, retries, ending balance
- Duplicate and cannibalization check result
- Live og:image verification result

4. Content Engines (9 Reference Engines)

The engine determines article structure, tone, visual approach, and product integration. Engine selection is **not a static lookup**. The pipeline uses a three-layer system: (1) the theme hint from the Excel queue, (2) SERP-driven validation during Step 5-7, and (3) dynamic engine composition for hybrid keywords. Reference: docs/ARTICLE_ENGINES_PLAYBOOK.md

Adaptive Engine Selection

Instead of blindly mapping theme to engine, the agent **validates or overrides** the theme hint based on what Google is actually ranking. If the top 5 SERP results for 'pearl necklace set' are all design listicles but the theme says 'Education', the agent should write a design listicle.

Layer	What It Does	When It Runs
1. Theme Hint	Initial engine suggestion from the Excel queue 'theme' column	At prompt build time (orchestrator)
2. SERP Validation	Agent scrapes top 5 Google results and checks if the dominant format matches the hint	Step 5-7 (agent)
3. Dynamic Composition	For hybrid keywords, agent blends a primary engine (70-80%) with a secondary engine (20-30%)	Step 7 (agent)

All 9 Reference Engines

#	Engine	Triggered By	Core Structure
1	Gift Guide	'gifts for sister', 'rakhi gifts', 'birthday gift for wife'	Recipient segments, occasion guide, 5-6 named SKU recommendations with gifting reasons
2	Buying Guide / Education	'how to check gold purity', '916 hallmark', 'GST on gold', 'bangle size'	Technical explainer, step-by-step, regulatory context, buyer implications, fact-check gate
3	Design Listicle	'gold ring designs', 'latest earrings', 'trendy necklace designs'	Style categories, trend analysis, outfit pairing, 'best for' context mapping
4	Wishes and Quotes Compilation	'happy diwali wishes', 'birthday wishes for crush', 'eid mubarak messages'	120+ curated wish/quote lines by relationship and tone, soft jewellery gift tie-in section
5	Comparison / Versus	'gold vs platinum', 'diamond vs moissanite', '22k vs 24k gold'	Quick verdict, Option A/B deep-dives, head-to-head comparison (bullet lists, NOT tables)
6	Cultural Significance / Heritage	'significance of mangalsutra', 'history of Polki', 'why do we wear toe rings'	Historical origins, spiritual meaning, regional variations, modern interpretations
7	Care and Maintenance How-To	'how to clean gold at home', 'how to store diamonds', 'jewellery polish tips'	Materials list, numbered step-by-step, mistakes to avoid, when to seek professional help
8	Trend Report / Seasonal Forecast	'jewellery trends 2027', 'summer jewellery trends', 'what is trending'	Macro trends, individual trend deep-dives, what is fading, adoption advice

#	Engine	Triggered By	Core Structure
9	Regional / Bridal Collection	'South Indian bridal jewellery', 'Bengali wedding jewellery', 'Maharashtrian'	Essential pieces per tradition, styling guide, regional accuracy, modern adaptations

H2 Priority Weighting (All Engines)

Priority	Source	Share	Placement
Spine	Volume-backed primary + supporting keywords from workbook row	~70-80%	Higher H2s + most FAQs
Parity	Competitor / SERP-consensus structural buckets	~20-30%	Lower H2s; keep shorter
Never	Soft parity buckets as separate URLs	0%	Stay on the single pillar URL

ENGINE OVERRIDE RULE: If the SERP analysis in Step 5 contradicts the theme hint from the workbook, the agent MUST follow the SERP (Google is showing what users want). The override and reasoning must be logged in the Final Report.

5. Image System (Types 1, 2, 3)

Type	What It Is	Source / Location	Usage in Article
Type 1 — Raw	Original studio jewellery photographs from BlueStone's photography sessions	ProductImages/raw/<category>/	Reference ONLY when writing Type 2/3 prompts. Never used directly in articles or carousel
Type 2 — AI Product	Styled product shots on plinths / draped fabric / warm-lit surfaces. 960×535 WebP	ProductImages/seo images/<Category>/<Product Name>.png	Mid-article 3D Coverflow carousel cards ONLY. Never as hero. Never plain white packshots
Type 3 — AI Concept	Lifestyle / atmosphere / editorial scenes generated by Higgsfield AI (nano_banana_pro, 16:9, 2K)	Generated during Step 12 via Higgsfield CLI	Hero (featured image) + 1-2 in-body editorial images. Never reuse hero in body

Type 3 Generation Style Lock

All Type 3 Higgsfield prompts MUST include this style language:

- Analog grain / subtle cinematic film grain
- Kodak Portra color science
- Highlight halation
- Creamy bokeh
- Filmic tonal response / highlight roll-off
- Editorial color grading
- Natural dynamic range / filmic contrast

Image SEO Requirements (Mandatory Every Publish)

Requirement	Rule
File format	WebP only for all published images. Type 3 preferred WebP for LCP/hero
Type 3 filename	{occasion}-{hero flatlay lifestyle}-{year}.webp
Type 2 filename	{product-slug}-carousel.webp
Alt text — hero	Primary keyword + year + brief scene/product description
Alt text — carousel	{occasion/KW + year} gift idea: The {Product Name}
Alt text — in-body Type 3	Unique keyword-led description with product + scene + year
Media library title	Descriptive (NOT IMG_1234 or bare slug). E.g. 'Valeria Rose Pendant carousel — Eid 2027'
Gutenberg sizeSlug	Always 'full' — never 'large' with explicit width/height/loading attributes
Captions	Standard <figcaption>editorial caption</figcaption> — no extra classes

Requirement	Rule
Product image links	Every product image in body wrapped in

■■ JEWELLERY MATERIAL TRUTH: BlueStone is a fine gold and diamond jeweller. It does NOT sell silver. Never recolor BlueStone SKUs in image prompts. Never describe authentic gold/diamond jewellery as silver.

6. Carousel & Product Rules

Official 3D Coverflow Carousel Specification

The mid-article carousel MUST strictly use the template from templates/eid_carousel_6_snippet.html. It MUST be a SINGLE unified wp:html block. No exceptions.

Requirement	Rule
Template source	templates/eid_carousel_6_snippet.html — MUST use this template, not custom versions
Block structure	ONE wp:html block containing: <style> + <div class='bs-cf' id='bs-cf-{slug}'> + <script>
CSS classes REQUIRED	.bs-cf-stage, .bs-cf-card, .bs-cf-prev, .bs-cf-next, .bs-cf-dots, .bs-cf-cta, is-pos-0 through is-pos--1
CSS values REQUIRED	.bs-cf-stage { height: 360px }, margin: 1.75rem auto 1.25rem, .bs-cf-card { width: min(420px, 78vw) }
NEVER use	bs-cf-wrap, bs-cf-track, or any unstyled custom wrapper classes
NEVER do	Omit <style> or <script>, split into multiple blocks, wrap lines in single quotes
Product count	Exactly 6 products per carousel
Image verification	ALL 6 image URLs verified HTTP 200 before insertion — MANDATORY
Card structure	Each card: + name + Buy now
After carousel	Follow immediately with ONLY a single 'Curated Design Highlights' paragraph. NEVER a repetitive product bullet list
Carousel position	Mid-article — NEVER the last block. Content continues after carousel

Product Selection Rules

- Source: Seo Products - consolidated.csv — PDP links, categories, SKU codes
- Count: 5-6 products per article
- Match products to article intent, theme, and keyword category
- Track rotation in output/product_rotation.json — avoid reusing same SKUs too frequently
- NO prices anywhere — not in body text, carousel cards, captions, or FAQs
- Gift guides: named product recommendations in BODY (not just carousel) with gifting reasons
- Buying guides: product placement secondary and useful (educational context, not forced gifting)
- Design listicles: products as design references with 'best for' style/context mapping

7. WordPress Publishing & Yoast SEO

API Configuration

Parameter	Value
Base URL	https://blog.bluestone.com/wp-json/wp/v2/
Authentication	WP_USER + WP_APP_PASSWORD (environment variables — never commit to git)
WP Username	blogbluestone
Author (Week 9)	Satyam — WP user ID 270271337
Author byline	By Satyam, BlueStone Editorial
BlogPosting schema author	Satyam
Theme	Creatio — single template (wp_id 29900)
Post title rendering	Theme renders WP post title as <h1> above featured image
H1 rule	ONE h1 only — the theme post title. NEVER inject a duplicate h1 in content

WordPress Category Mapping

Article Intent / Engine	WordPress Category IDs
General gift guide	Gift (554493424)
Seasonal / festival gift guide	Gift (554493424) + Festive Wishes (554493477)
Wedding / couple jewellery	Gift (554493424) + Wedding Jewellery (554493443)
Gold buying / purity guide	Gold (554493348) + Jewellery Problem and Solution (554493465)
Design listicle / style discovery	Jewellery Trends (554493317) or Jewellery and Lifestyle (554493425)

Yoast SEO Fields

Field	Value / Rule
_yoast_wpseo_focuskw	Primary keyword from workbook row
_yoast_wpseo_title	CTR-optimized title with primary keyword — max 60 chars preferred
_yoast_wpseo_metadesc	Compelling 150-160 character description accurately summarizing article
Featured image (og:image)	Hero Type 3 image — set as both featured_media and Yoast social image

Field	Value / Rule
Social image fallback	If Yoast social-image field not exposed via REST: use authenticated editor UI for that field only

Strict Gutenberg Block Syntax Rules

Every WordPress block comment MUST be strictly formatted with opening and closing hyphens: `<!-- wp:paragraph --> ... <!-- /wp:paragraph -->`. NEVER emit: `<!-- /wp:paragraph>` (missing `--`) or `<!-- /wp:paragraph)` (parenthesis typo). Every `wp:paragraph` block MUST explicitly enclose text in `<p>...</p>` tags.

8. Checkpoint & Idempotency System

Each rank has an atomic JSON checkpoint at `output/checkpoints/week9_rank{N}.json`. Steps are marked atomically with file-system locking. If the agent crashes, restarting the same command skips completed steps and resumes from the last pending one.

Step Names (Exact Order)

#	Step Name	Checkpoint Key	Phase
1	Read Docs	read_docs	Phase 1
2	Verify Higgsfield	verify_higgsfield	Phase 1
3	Inspect Row	inspect_row	Phase 1
4	Duplicate Check	duplicate_check	Phase 1
5	SERP Intelligence	serp_intelligence	Phase 2
6	Fact Check	fact_check	Phase 2
7	Keyword Map	keyword_map	Phase 3
8	Structure & Visuals	structure_visuals	Phase 3
9	Draft Article	draft	Phase 4
10	Product & Carousel	product_media	Phase 4
11	Publish WordPress	publish_wordpress	Phase 5
12	Generate Type 3	generate_type3	Phase 5
13	Patch Media	patch_type3	Phase 5
14	Live QA	live_qa	Phase 6
15	Update Status	update_status	Phase 6
16	Final Report	final_report	Phase 6

Step Status Values

Status	Meaning
pending	Step has not started yet
running	Step is currently in progress
done	Step completed successfully — artifact or external ID verified
failed	Step failed due to agent or script error
skipped	Step intentionally not applicable for this run

Checkpoint Commands

```
# View checkpoint state
python3 -B scripts/article_checkpoint.py show --path output/checkpoints/week9_rank{N}.json

# Mark a step done
python3 -B scripts/article_checkpoint.py mark --path <path> --step <step_name> --status done --detail key=value

# Initialize checkpoint
python3 -B scripts/article_checkpoint.py init --path <path> --rank <N> --slug <slug> --primary "<keyword>"
```

Pipeline Outcomes

Outcome	Checkpoint Status	Meaning
published	complete	Post published, verified, and live QA passed
skipped_existing_intent	complete	Duplicate intent detected — no post or media created
incomplete	incomplete	Agent exited successfully but live QA did not pass
failed	failed	Agent or script error caused step failure

9. Mandatory Content Gates (Hard Rules)

These are non-negotiable rules enforced in the pipeline prompt. Violation causes QA failure or produces broken/non-compliant articles. Review these when making any workflow changes.

Article Content Rules

- Action must be New — skip Done and Skip-Existing rows without generating content
- Duplicate intent check MUST run before any research or drafting
- SERP intelligence (serp_intelligence step) runs AFTER duplicate check, BEFORE keyword mapping
- All 5 SERP URLs must be scraped and read in full — no skipping
- Competitor URL analyzed when present — never invent a competitor URL if blank
- Supporting keywords audited: irrelevant phrases explicitly REJECTED with reasons listed
- Every article uses the appropriate engine from ARTICLE_ENGINES_PLAYBOOK.md
- Gift guides: 5-6 named BlueStone product recommendations in the BODY (not just carousel)
- Buying guides: explain calculations, buyer implications, regulatory context clearly
- Factual topics (GST, hallmarking, purity, regulations): browse current authoritative sources, cite them
- NO prices anywhere in the article
- NO fake statistics
- NO unsupported legal or tax claims
- NO competitor criticism
- NO em dashes (—), en dashes (–), or spaced hyphens (word - word)
- Author: Satyam (ID 270271337), byline: 'By Satyam, BlueStone Editorial'
- NO HTML tables — use structured bullet lists, cards, or styled PNG graphics instead
- Jewellery material: BlueStone sells gold and diamond ONLY — no silver
- EXACTLY ONCE content: no duplicate sections, no repeated H2s, no concatenated article halves
- Article depth preserved even when image generation runs in parallel
- 120+ distinct list lines for quotes/captions/listicle articles (unless row specifies different format)

Section Ordering Rules (MANDATORY)

1. Intro → 2. Body H2s → 3. Carousel (mid-article) → 4. More Body H2s → 5. Related Guides section (4-5 internal links) → 6. Final Thoughts / Conclusion → 7. Frequently Asked Questions (5-7 visible Q&As) → 8. JSON-LD schema Final Thoughts MUST appear BEFORE FAQs. FAQs MUST be the last content section before JSON-LD. Related Guides section MUST appear between Conclusion and FAQs.

Synchronous Execution Gate

CRITICAL: The AI agent **MUST** execute **ALL** commands synchronously. **NO** background tasks, **NO** async delays. Every python script, curl call, Higgsfield generation, and WordPress upload runs in the foreground and blocks until full completion. The agent **NEVER** ends its turn while a task is running.

10. CLI Reference & How to Run

Standard Production Command

```
python3 -B scripts/article_pipeline.py produce \  
--ranks 69 \  
--manifest output/week9_69.json \  
--timers \  
--approve-for-me
```

Auto-pick Next Unfinished Rank

```
python3 -B scripts/article_pipeline.py produce \  
--next \  
--manifest output/week9_next.json \  
--timers --approve-for-me
```

All CLI Options

Flag	Purpose	Notes
--ranks N or N-M	Specific rank(s) to generate	e.g. --ranks 69 or --ranks 69-73 or --ranks 69,72,75
--next [COUNT]	Auto-pick next N unfinished Action=New ranks	Default COUNT=1 if omitted
--manifest <path>	Run manifest JSON path (audit trail)	REQUIRED — used for status command too
--timers	Enable step-by-step timing display in terminal	Recommended for monitoring
--approve-for-me	Auto-approve agent workspace writes	Needed for non-interactive runs
--dry-run	Build prompt only — do not launch agent	For prompt inspection and testing
--verbose	Stream full agent output to terminal	Full log always saved regardless
--restart	Archive old checkpoint and restart selected ranks from step 1	Requires explicit --ranks
--allow-existing-intent	Override duplicate-intent gate after documenting the conflict	Requires --restart
--force	Rerun already-completed ranks	Use for intentional controlled reruns only
--runner agy	Use AGY agent (default) or codex	Default: agy
--model <name>	Specify AI model for AGY runner	Optional

Flag	Purpose	Notes
--continue-on-error	Continue to next rank even if current rank fails	For batch runs
--heartbeat-seconds N	Seconds between progress updates (min 10)	Default: 30

Status Command

```
# Check run status
python3 -B scripts/article_pipeline.py status --manifest output/week9_69.json

# JSON output
python3 -B scripts/article_pipeline.py status --manifest output/week9_69.json --json
```

Expected Terminal Output (with --timers)

```
=====
Week 9 Rank 69: light earrings (Runner: agy)
Engine: buying_guide
=====
step 1: read_docs | completed in 1m 12s
step 2: verify_higgsfield | completed in 21s
step 3: inspect_row | completed in 23s
step 4: duplicate_check | completed in 1m 0s
step 5: serp_intelligence | completed in 2m 30s
step 6: fact_check | completed in 13s
step 7: keyword_map | completed in 45s
step 8: structure_visuals | completed in 28s
step 9: draft | completed in 8m 15s
step 10: product_media | completed in 3m 10s
step 11: publish_wordpress | completed in 2m 5s
step 12: generate_type3 | completed in 5m 30s
step 13: patch_type3 | completed in 1m 45s
step 14: live_qa | completed in 2m 0s
step 15: update_status | completed in 15s
step 16: final_report | completed in 24s
```

11. Error Recovery Playbook

Scenario	Recovery Action	Command / Notes
Agent crashes mid-step	Rerun the same command — checkpoint skips all completed steps and resumes from last pending	<code>python3 -B scripts/article_pipeline.py produce --ranks N --manifest <same manifest> --timers --approve-for-me</code>
Want clean restart from step 1	Archive old checkpoint and restart	<code>--restart</code> flag (requires explicit <code>--ranks</code>). Old checkpoint archived to <code>output/checkpoints/archive/</code>
Duplicate slug found in WordPress	Agent stops safely. Reconcile manually: decide if existing post is the same intent or if new slug is needed	May use <code>--allow-existing-intent</code> with <code>--restart</code> if user approves a differentiated rerun
Higgsfield generation fails	Agent waits 45 seconds and retries once automatically. If still failing, step is marked failed	Rerun with same command — <code>generate_type3</code> step will be retried from checkpoint
Duplicate search intent detected	Pipeline halts with <code>skipped_existing_intent</code> outcome. No post or media created	Review the conflicting URL. If differentiation is possible, use <code>--allow-existing-intent --restart</code>
Carousel is broken on live site	Live QA step detects this and immediately replaces with <code>eid_carousel_6_snippet.html</code> template	If QA detects unstyled <code>bs-cf-wrap</code> or broken images, carousel is patched before QA passes
Incomplete checkpoint from previous run	Rerun the command — checkpoint automatically resumes from last pending step	Use <code>--restart</code> only if you want to wipe and start fresh from step 1
Status CSV corrupted or headerless	Pipeline auto-detects legacy format and migrates rows to new schema	<code>upsert_status_record()</code> handles both legacy 8-column and current header formats
Another pipeline instance is running	File lock prevents concurrent runs. Wait for the running pipeline to finish	Lock file: <code>output/.locks/article_pipeline.lock</code>

12. Monitoring & Status Tracking

Status CSV — output/Week9_Blog_Queue_status.csv

Column	Description
rank	Integer rank from Week 9 workbook
primary	Primary keyword
slug	Published URL slug
blog_url	Full published URL on blog.bluestone.com
wp_post_id	WordPress post ID
status	published skipped_existing_intent failed
carousel_media	WordPress media IDs for carousel images (JSON array)
type3_media	WordPress media IDs for Type 3 images (JSON array)
lines	Number of section lines in the draft
visible_words	Verified visible word count from live QA
notes	Run notes: engine, checkpointed components, QA result

Manifest JSON — output/week9_{ranks}.json

- Created per run — contains all run metadata, timing, exit codes, outcomes, per-rank details
- Fields: pipeline, created_at, workspace, week9_workbook, week9_csv, requested_ranks, runs[]
- Each run entry: rank, primary, slug, engine, status, started_at, finished_at, duration_seconds, exit_code, outcome, prompt_path, log_path, final_path, checkpoint_path

Checkpoint JSON — output/checkpoints/week9_rank{N}.json

- Created per rank — tracks per-step status, timestamps, and key artifacts
- Fields: pipeline, rank, slug, primary, created_at, status, outcome, steps{}
- Each step entry: status (pending/running/done/failed/skipped), started_at, finished_at, detail{}
- detail{} contains step-specific data: wp_post_id, live_url, media_ids, word_count, result, etc.
- Archive on restart: moved to output/checkpoints/archive/week9_rank{N}_{timestamp}.json

SERP Cache — output/serp_cache/

- Stores pre-fetched Semrush SERP data per keyword
- If cached data exists for the primary keyword, it is injected directly into the prompt
- If no cache: agent searches Google India live during Step 5

- Cache managed by `scripts/serp_competitor_intel.py`

13. Key Files Quick Reference

File	Purpose	When to Read
HANDOFF.md	Project context, current state, active queue	Always first
docs/SOP_ARTICLE_GENERATION.md	Primary execution runbook Step 0 → Step 10	Before any new run
docs/ARTICLE_WORKFLOW.md	Canonical hard rules: H1, images, carousel, dash rules	When reviewing rules
docs/ARTICLE_ENGINES_PLAYBOOK.OK.md	Engine selection (Gift/Buying/Design) and H2 strategy	During keyword mapping
docs/bluestone-blog-master-prompt-v5.md	Master article drafting prompt template	During draft step
docs/Blog-SEO-AEO-GEO-Checklist-v2.md	Pre-publish QA checklist (mandatory every publish)	Before live QA
docs/HIGGSFIELD_IMAGE_GENERATION.md	Higgsfield SOP, prompt templates, Image SEO table	During image generation
docs/competitor-blog-analysis-SKILL.md	Competitor analysis template and structure	During fact-check
cursor-rules/new-blogs-only.mdc	Rule: new posts only, never optimize old posts	If ever tempted to edit existing posts
cursor-rules/carousel-seo-images.mdc	Carousel image sourcing, conversion, pre-flight	During carousel building
cursor-rules/type3-fair-skinned-indians.mdc	Type 3 diversity/representation guidelines	During image prompt writing
cursor-rules/image-seo.mdc	Image SEO: alts, filenames, WebP, sizeSlug rules	Before media upload
cursor-rules/product-rotation-captions-education.mdc	Product rotation and educational caption rules	During product selection
templates/eid_carousel_6_snippet.html	Official 3D Coverflow carousel template (6 cards)	During carousel assembly
Seo Products - consolidated.csv	Product catalog: name, SKU, category, PDP URL	During product selection
output/Week9_Blog_Queue_status.csv	Terminal status tracker — which ranks are done	Before selecting next rank
output/product_rotation.json	SKU rotation state — which products used per rank	During product selection
SEO_Pipeline_Source_of_Truth.pdf (this file)	Complete pipeline documentation — source of truth	Any time workflow questions arise

BlueStone SEO Generation Pipeline — Source of Truth

Generated: September 29, 2026 at 15:20 IST | Pipeline Version: Week 9 with SERP Intelligence

This document is the authoritative reference. Update this PDF whenever the workflow changes.